Course Name: Geographic Information System

Week 06 - Assignment 06 - (Jul-2026)

TYPE OF QUESTIONS: MCQ/MSQ

Number of questions: 10 **Total marks: 10x1= 10**

QUESTION 1

GPS determines the location of a receiver using the principle of _____.

a) Triangulation
b) Trilateration
c) Reflection
d) Interpolation

Answer: (b)

Detailed Solution:

GPS works on the principle of trilateration, where distances from multiple satellites are used to determine the exact position of a receiver.

QUESTION 2

The GPS space segment mainly consists of _____ satellites.

a) 14
b) 21
c) 24
d) 32

Answer: (c)

Detailed Solution:

The standard GPS constellation consists of 24 operational satellites orbiting the Earth.

QUESTION 3

GPS signals are transmitted in the _____ region of the electromagnetic spectrum.

a) Visible
b) Infrared

NPTEL

Course Name: Geographic Information System

Week 06 - Assignment 06 - (Jul-2026)

c) Microwave
d) Ultraviolet

Answer: (c)

Detailed Solution:

GPS satellites transmit signals in the microwave region because microwaves can pass through clouds, rain, and fog with minimal interference.

QUESTION 4

The Indian Regional Navigation Satellite System (IRNSS) consists of _____ satellites.

a) 5
b) 6
c) 7
d) 8

Answer: (c)

Detailed Solution:

IRNSS, also known as NavIC is made up of 7 satellites providing regional navigation services around India.

QUESTION 5

Which GNSS system was developed by Russia?

a) Galileo
b) GPS
c) QZSS
d) GLONASS

Answer: (d)

Detailed Solution:

GLONASS is the satellite navigation system developed and operated by Russia.

NPTEL

Course Name: Geographic Information System

Week 06 - Assignment 06 - (Jul-2026)

QUESTION 6

Errors in GPS signals caused due to reflection from nearby buildings are known as _____.

a) Ephemeris errors
b) Multipath errors
c) Atmospheric delay
d) Receiver noise

Answer: (b)

Detailed Solution:

Multipath errors occur when GPS signals bounce off buildings or other surfaces before reaching the receiver.

QUESTION 7

The largest source of GPS positioning error is generally caused by _____.

a) Receiver noise
b) Multipath
c) Ionospheric delay
d) Satellite geometry

Answer: (c)

Detailed Solution:

Charged particles in the ionosphere delay GPS signals, causing significant positioning errors.

QUESTION 8

WAAS is mainly used to improve _____.

a) Satellite speed
b) Navigation accuracy
c) Map projection
d) Data storage

Answer: (b)

NPTEL

Course Name: Geographic Information System

Week 06 - Assignment 06 - (Jul-2026)

Detailed Solution:

WAAS (Wide Area Augmentation System) improves GPS navigation accuracy and reliability, especially for aviation purposes.

QUESTION 9

The GPS master control station is located in _____.

a) Nevada
b) Los Angeles
c) Colorado Springs
d) Washington DC

Answer: (c)

Detailed Solution:

The GPS master control station responsible for monitoring and controlling GPS satellites is located at Colorado Springs, USA.

QUESTION 10

Selective Availability (SA) in GPS was officially turned off in the year _____.

a) 1995
b) 1998
c) 2000
d) 2005

Answer: (c)

Detailed Solution:

Selective Availability, which intentionally reduced GPS accuracy for civilian users, was discontinued in 2000 to allow improved positioning accuracy.

NPTEL